

Programming Fundamentals

Assignment 02

Codes

Program 01

```
#include<iostream>

using namespace std;

string customerName;

int customerID, customerType, meterType;

void registerCustomer()
{
    cout << "Enter Customer Name: ";

    cin >> customerName;

    cout << "Enter Customer ID: ";

    cin >> customerID;

    cout << "1. Household\n2. Commercial\n";

    cout << "Enter Customer Type: ";

    cin >> customerType;

    cout << "1. First Meter\n2. Second Meter\n";

    cout << "Enter Meter Type: ";

    cin >> meterType;
}

double calculateConsumption(int units)
{
    double rates[8] = {12.21, 14.53, 31.51, 38.41, 41.62, 43.04, 44.18, 49.10};
```

```
    if(units<= 100)
        return units * rates[0];
    else if(units<= 200)
        return units * rates[1];
    else if(units<= 300)
        return units * rates[2];
    else if(units<= 400)
        return units * rates[3];
    else if(units<= 500)
        return units * rates[4];
    else if(units<= 600)
        return units * rates[5];
    else if(units<= 700)
        return units * rates[6];
    else
        return units * rates[7];
}

double calculateGST(double amount)
{
    return amount * 0.18;
}

double calculateIncomeTax(double amount)
{
    if(customerType == 1)
        return amount * 0.10;
```

```
    else
        return amount * 0.15;
}

double calculateDuty(double amount)
{
    return amount * 0.015;
}

double calculateFixedCharges(int units)
{
    if(units >= 301 && units <= 400)
        return 200;
    else if(units >= 401 && units <= 500)
        return 400;
    else if(units >= 501 && units <= 600)
        return 600;
    else if(units >= 601 && units <= 700)
        return 800;
    else if(units > 700)
        return 1000;
    else
        return 0;
}

void calculateBill()
{
    int units;
```



```
cout << "Enter Units Consumed: ";

cin >> units;

double consumption = calculateConsumption(units);

double duty = calculateDuty(consumption);

double gst = calculateGST(consumption);

double incomeTax = calculateIncomeTax(consumption);

double fixedCharges = calculateFixedCharges(units);

double meterRent = 250;

double tvFee = 35;

double totalBill = consumption + duty + gst +
    incomeTax + fixedCharges +
    meterRent + tvFee;

cout << "\n===== LESCO BILL =====\n";

cout << "Customer Name: " << customerName << endl;

cout << "Customer ID: " << customerID << endl;

cout << "Units Consumed: " << units << endl;

cout << "Consumption Charges: Rs. " << consumption << endl;

cout << "Electricity Duty: Rs. " << duty << endl;

cout << "Fixed Charges: Rs. " << fixedCharges << endl;

cout << "Meter Rent: Rs. " << meterRent << endl;

cout << "TV Fee: Rs. " << tvFee << endl;

cout << "GST: Rs. " << gst << endl;

cout << "Income Tax: Rs. " << incomeTax << endl;

cout << "Total Bill: Rs. " << totalBill << endl;

cout << "=====\\n";
```

```
}
```

```
void newConnection()
```

```
{
```

```
    double charges = 0;
```

```
    if(customerType == 1)
```

```
    {
```

```
        if(meterType == 1)
```

```
            charges = 2500;
```

```
        else
```

```
            charges = 5000;
```

```
    }
```

```
    else
```

```
    {
```

```
        if(meterType == 1)
```

```
            charges = 35000;
```

```
        else
```

```
            charges = 70000;
```

```
    }
```

```
    charges += 250000;
```

```
    cout << "\nNew Connection Charges: Rs. "
```

```
        << charges << endl;
```

```
}
```

```
void displayCustomer()
```

```
{
```

```
    cout << "\nCustomer Name: " << customerName << endl;
```

```
cout << "Customer ID: " << customerID << endl;

if(customerType == 1)
    cout << "Customer Type: Household" << endl;
else
    cout << "Customer Type: Commercial" << endl;
}

int main()
{
    registerCustomer();

    int choice;

    do
    {
        cout << "\n===== MAIN MENU =====\n";
        cout << "1. Calculate Bill\n";
        cout << "2. New Connection\n";
        cout << "3. View Customer Details\n";
        cout << "4. Exit\n";
        cout << "Enter Choice: ";
        cin >> choice;
        switch(choice)
        {
            case 1:
                calculateBill();
                break;
            case 2:
```



```
        newConnection();
        break;
    case 3:
        displayCustomer();
        break;
    case 4:
        cout << "Program Ended\n";
        break;
    default:
        cout << "Invalid Choice\n";
    }
} while(choice != 4);
return 0;
}
```

Program 02

```
#include<iostream>

using namespace std;

string customerName, contactNumber;

int orderType, persons;

string foodItems[8] = {
    "Chicken Burger",
    "Zinger Burger",
    "Pizza Small",
    "Pizza Large",
```

```
    "Chicken Biryani",  
    "BBQ Platter",  
    "Fries",  
    "Cold Drink"  
};
```

```
double prices[8] = {  
    450, 550, 900, 1800,  
    350, 1200, 250, 120  
};
```

```
double foodBill = 0;
```

```
void registerCustomer()  
{  
    cout << "Enter Customer Name: ";  
    cin >> customerName;  
  
    cout << "Enter Contact Number: ";  
    cin >> contactNumber;  
  
    cout << "1. Dine In\n2. Take Away\n";  
    cout << "Enter Order Type: ";  
    cin >> orderType;  
  
    cout << "Enter Number of Persons: ";
```



```
    cin >> persons;
}

void displayMenu()
{
    cout << "\n===== FOOD MENU =====\n";

    for(int i=0; i<8; i++)
    {
        cout << i+1 << ". "
            << foodItems[i]
            << " - Rs. "
            << prices[i] << endl;
    }
}

void placeOrder()
{
    int itemNo, quantity;

    cout << "Enter Item Number: ";
    cin >> itemNo;

    cout << "Enter Quantity: ";
    cin >> quantity;
```

```
foodBill = foodBill + (prices[itemNo-1] * quantity);
```

```
cout << "Item Added Successfully\n";
```

```
}
```

```
double calculateServiceCharges()
```

```
{
```

```
if(orderType == 1)
```

```
    return foodBill * 0.10;
```

```
else
```

```
    return foodBill * 0.05;
```

```
}
```

```
double calculateGST()
```

```
{
```

```
    return foodBill * 0.16;
```

```
}
```

```
double calculateDiscount()
```

```
{
```

```
if(foodBill >= 3000 && foodBill <= 5000)
```

```
    return foodBill * 0.05;
```

```
else if(foodBill > 5000 && foodBill <= 10000)
```

```
    return foodBill * 0.10;
```

```
else if(foodBill > 10000)
    return foodBill * 0.15;

return 0;
}

void calculateBill()
{
    double serviceCharges = calculateServiceCharges();
    double gst = calculateGST();
    double discount = calculateDiscount();

    double total =
    foodBill + serviceCharges + gst - discount;

    cout << "\n===== RESTAURANT BILL =====\n";
    cout << "Customer Name: " << customerName << endl;
    cout << "Food Bill: Rs. " << foodBill << endl;
    cout << "Service Charges: Rs. "
        << serviceCharges << endl;
    cout << "GST: Rs. " << gst << endl;
    cout << "Discount: Rs. "
        << discount << endl;

    if(total>5000)
        cout << "Free Delivery Applied\n";
```



```
cout << "Total Payable: Rs. "
    << total << endl;

cout << "=====\n";
}

void customerDetails()
{
    cout << "\nCustomer Name: "
        << customerName << endl;

    cout << "Contact Number: "
        << contactNumber << endl;

    cout << "Persons: "
        << persons << endl;
}

int main()
{
    registerCustomer();

    int choice;

    do
```

```
{  
  
    cout << "\n===== MAIN MENU =====\n";  
  
    cout << "1. View Food Menu\n";  
  
    cout << "2. Place Order\n";  
  
    cout << "3. Calculate Bill\n";  
  
    cout << "4. View Customer Details\n";  
  
    cout << "5. Exit\n";  
  
  
    cout << "Enter Choice: ";  
  
    cin >> choice;  
  
  
    switch(choice)  
    {  
  
        case 1:  
  
            displayMenu();  
  
            break;  
  
  
        case 2:  
  
            placeOrder();  
  
            break;  
  
  
        case 3:  
  
            calculateBill();  
  
            break;  
  
  
        case 4:
```

```
        customerDetails();

        break;

    case 5:

        cout << "Program Ended\n";

        break;

    default:

        cout << "Invalid Choice\n";

    }

} while(choice != 5);

return 0;

}
```

Program 03

```
#include<iostream>

using namespace std;

string customerName;

int customerID, customerType, paymentMethod;

string items[8] = {
```



```
"Rice",  
"Sugar",  
"Cooking Oil",  
"Milk Pack",  
"Tea Pack",  
"Flour",  
"Eggs",  
"Detergent"  
};
```

```
double prices[8] = {  
    350,180,580,220,  
    450,950,320,600  
};
```

```
double grossBill = 0;
```

```
void registerCustomer()  
{  
    cout << "Enter Customer Name: ";  
    cin >> customerName;  
  
    cout << "Enter Customer ID: ";  
    cin >> customerID;
```

```
cout << "1. Regular Customer\n2. Member Customer\n";
```

```
cout << "Enter Customer Type: ";

cin >> customerType;


cout << "1. Cash\n2. Card\n";

cout << "Enter Payment Method: ";

cin >> paymentMethod;
}


void displayItems()
{
    cout << "\n===== GROCERY ITEMS =====\n";

    for(int i=0; i<8; i++)
    {
        cout << i+1 << ". "

            << items[i]

            << " - Rs. "

            << prices[i] << endl;
    }
}


void addToCart()
{

    int itemNo, quantity;


    cout << "Enter Item Number: ";
```



```
cin >> itemNo;
```

```
cout << "Enter Quantity: ";
```

```
cin >> quantity;
```

```
grossBill += prices[itemNo-1] * quantity;
```

```
cout << "Item Added Successfully\n";
```

```
}
```

```
double calculateTax()
```

```
{
```

```
    return grossBill * 0.05;
```

```
}
```

```
double membershipDiscount()
```

```
{
```

```
    if(customerType == 2)
```

```
        return grossBill * 0.07;
```

```
    return 0;
```

```
}
```

```
double billDiscount()
```

```
{
```

```
    if(grossBill >= 5000 && grossBill <= 10000)
```

```
    return grossBill * 0.05;
```

```
else if(grossBill > 10000)
```

```
    return grossBill * 0.10;
```

```
return 0;
```

```
}
```

```
double cardCharges()
```

```
{
```

```
    if(paymentMethod == 2)
```

```
        return grossBill * 0.02;
```

```
return 0;
```

```
}
```

```
void calculateBill()
```

```
{
```

```
    double tax = calculateTax();
```

```
    double memberDis = membershipDiscount();
```

```
    double billDis = billDiscount();
```

```
    double cardCharge = cardCharges();
```

```
    double total =
```

```
        grossBill + tax + cardCharge
```

```
        - memberDis - billDis;
```

```
int generatedPoints = total / 100;
```

```
int oldPoints;
```

```
cout << "Enter Existing Loyalty Points: ";
```

```
cin >> oldPoints;
```

```
int totalPoints = generatedPoints + oldPoints;
```

```
int choice;
```

```
cout << "Press 1 to Redeem Points\n";
```

```
cout << "Press 2 to Continue\n";
```

```
cin >> choice;
```

```
if(choice == 1)
```

```
{
```

```
    total = total - totalPoints;
```

```
}
```

```
cout << "\n===== GROCERY BILL =====\n";
```

```
cout << "Customer Name: "
```

```
    << customerName << endl;
```

```
cout << "Gross Bill: Rs. "
```

```
    << grossBill << endl;
```



```
cout << "Sales Tax: Rs. "
```

```
<< tax << endl;
```

```
cout << "Member Discount: Rs. "
```

```
<< memberDis << endl;
```

```
cout << "Bill Discount: Rs. "
```

```
<< billDis << endl;
```

```
cout << "Card Charges: Rs. "
```

```
<< cardCharge << endl;
```

```
cout << "Loyalty Points: "
```

```
<< totalPoints << endl;
```

```
cout << "Total Payable: Rs. "
```

```
<< total << endl;
```

```
cout << "=====\n";
```

```
}
```

```
void customerDetails()
```

```
{
```

```
cout << "\nCustomer Name: "
```

```
<< customerName << endl;
```

```
    cout << "Customer ID: "
        << customerID << endl;
}

int main()
{
    registerCustomer();

    int choice;

    do
    {
        cout << "\n===== MAIN MENU =====\n";
        cout << "1. View Grocery Items\n";
        cout << "2. Add Items To Cart\n";
        cout << "3. Calculate Bill\n";
        cout << "4. View Customer Details\n";
        cout << "5. Exit\n";

        cout << "Enter Choice: ";
        cin >> choice;

        switch(choice)
        {
            case 1:
                displayItems();
```

break;

case 2:

addToCart();

break;

case 3:

calculateBill();

break;

case 4:

customerDetails();

break;

case 5:

cout << "Program Ended\n";

break;

default:

cout << "Invalid Choice\n";

}

}while(choice != 5);

return 0;

}

Program 04

```
#include<iostream>
```

```
#include<string>
```

```
using namespace std;
```

```
string userName, email, city, customerType;
```

```
int paymentMethod;
```

```
string products[8] = {
```

```
    "T-Shirt",
```

```
    "Jeans",
```

```
    "Shoes",
```

```
    "Watch",
```

```
    "Handbag",
```

```
    "Headphones",
```

```
    "Mobile Cover",
```

```
    "Perfume"
```

```
};
```

```
float prices[8] = {1200,3500,5000,2500,4200,3000,700,2800};
```

```
float productTotal = 0;
```

```
void registerUser()
```

```
{
```

```
    cin.ignore();
```

```
cout << "Enter User Name: ";  
getline(cin, userName);
```

```
cout << "Enter Email: ";  
getline(cin, email);
```

```
cout << "Enter City: ";  
getline(cin, city);
```

```
cout << "Customer Type (New/Returning): ";  
getline(cin, customerType);
```

```
cout << "Payment Method\n";  
cout << "1. Cash on Delivery\n";  
cout << "2. Debit/Credit Card\n";  
cout << "Enter Choice: ";  
cin >> paymentMethod;
```

```
}
```

```
void displayProducts()
```

```
{
```

```
    cout << "\n===== PRODUCT LIST =====\n";
```

```
    for(int i=0;i<8;i++)
```

```
{
```



```
        cout << i+1 << ". "  
        << products[i]  
        << " - Rs. "  
        << prices[i]  
        << endl;  
    }  
}
```

```
void addToCart()
```

```
{
```

```
    int productNo, quantity;
```

```
    cout << "\nEnter Product Number: ";
```

```
    cin >> productNo;
```

```
    cout << "Enter Quantity: ";
```

```
    cin >> quantity;
```

```
    if(productNo>=1 && productNo<=8)
```

```
{
```

```
    productTotal += prices[productNo-1] * quantity;
```

```
    cout << "Product Added Successfully!\n";
```

```
}
```

```
else
```

```
{
```

```
    cout << "Invalid Product Number!\n";
```

```
    }  
}
```

```
float calculateGST()  
{  
    return productTotal * 0.17;  
}
```

```
float deliveryCharges()  
{  
    if(city=="Lahore" || city=="Karachi" || city=="Islamabad")  
        return 250;  
  
    return 500;  
}
```

```
float customerDiscount()  
{  
    if(customerType=="New" || customerType=="new")  
        return productTotal * 0.05;  
  
    return productTotal * 0.10;  
}
```

```
float orderDiscount()  
{
```

```
if(productTotal>=5000 && productTotal<=10000)
```

```
    return productTotal * 0.05;
```

```
else if(productTotal>10000)
```

```
    return productTotal * 0.12;
```

```
return 0;
```

```
}
```

```
float paymentCharges()
```

```
{
```

```
    if(paymentMethod==2)
```

```
        return productTotal * 0.025;
```

```
return 0;
```

```
}
```

```
void checkoutBill()
```

```
{
```

```
    float gst = calculateGST();
```

```
    float delivery = deliveryCharges();
```

```
    float customerDisc = customerDiscount();
```

```
    float orderDisc = orderDiscount();
```

```
    float paymentFee = paymentCharges();
```

```
float finalBill =
```


productTotal +

gst +

delivery +

paymentFee -

customerDisc -

orderDisc;

cout << "\n===== ONLINE SHOPPING BILL =====\n";

cout << "User Name: "

<< userName << endl;

cout << "City: "

<< city << endl;

cout << "Customer Type: "

<< customerType << endl;

cout << "\nProduct Total: Rs. "

<< productTotal;

cout << "\nGST: Rs. "

<< gst;

cout << "\nDelivery Charges: Rs. "

<< delivery;

```
cout << "\nCustomer Discount: Rs. "
    << customerDisc;

cout << "\nOrder Discount: Rs. "
    << orderDisc;

cout << "\nPayment Charges: Rs. "
    << paymentFee;

cout << "\n-----";

cout << "\nFinal Payable Amount: Rs. "
    << finalBill;

cout << "\nThank You for Shopping :)";

cout << "\n=====\\n";
}

void userDetails()
{
    cout << "\nUser Name: "
        << userName;

    cout << "\nEmail: "
```

```
<< email;
```

```
cout << "\nCity: "
```

```
<< city;
```

```
cout << "\nCustomer Type: "
```

```
<< customerType << endl;
```

```
}
```

```
int main()
```

```
{
```

```
    registerUser();
```

```
    int choice;
```

```
    do
```

```
    {
```

```
        cout << "\n===== ONLINE SHOPPING MENU =====\n";
```

```
        cout << "1. View Products\n";
```

```
        cout << "2. Add Product To Cart\n";
```

```
        cout << "3. Calculate Checkout Bill\n";
```

```
        cout << "4. View User Details\n";
```

```
        cout << "5. Exit\n";
```

```
        cout << "Enter Choice: ";
```

```
        cin >> choice;
```



```
switch(choice)
{
    case 1:
        displayProducts();
        break;

    case 2:
        addToCart();
        break;

    case 3:
        checkoutBill();
        break;

    case 4:
        userDetails();
        break;

    case 5:
        cout << "Program Ended.\n";
        break;

    default:
        cout << "Invalid Choice!\n";
}
```

```
    }while(choice!=5);

    return 0;
}
```

Program 05

```
#include<iostream>

#include<string>

using namespace std;

string clientName, businessName, businessType;

int campaignDays;

int selectedPlatform = 0;

string platforms[3] = {

    "Instagram",

    "Facebook",

    "LinkedIn"

};

float platformCharges[3] = {

    15000,

    12000,
```


20000

};

void registerClient()

{

cin.ignore();

cout << "Enter Client Name: ";

getline(cin, clientName);

cout << "Enter Business Name: ";

getline(cin, businessName);

cout << "Business Type (Small/Medium/Corporate): ";

getline(cin, businessType);

cout << "Campaign Duration (Days): ";

cin >> campaignDays;

}

void displayPlatforms()

{

cout << "\n===== SOCIAL MEDIA PLATFORMS =====\n";

for(int i=0; i<3; i++)

{

```
        cout << i+1 << ". "  
        << platforms[i]  
        << " - Rs. "  
        << platformCharges[i]  
        << endl;  
    }  
}
```

```
void selectPlatform()  
{  
    cout << "\nSelect Platform (1-3): ";  
    cin >> selectedPlatform;  
  
    if(selectedPlatform < 1 || selectedPlatform > 3)  
    {  
        cout << "Invalid Selection!\n";  
        selectedPlatform = 0;  
    }  
    else  
    {  
        cout << "Platform Selected Successfully!\n";  
    }  
}
```

```
float calculatePostDesignCost()  
{
```

```
int staticPosts, reelPosts, carouselPosts;
```

```
cout << "\nEnter Number of Static Posts: ";
```

```
cin >> staticPosts;
```

```
cout << "Enter Number of Reel/Video Posts: ";
```

```
cin >> reelPosts;
```

```
cout << "Enter Number of Carousel Posts: ";
```

```
cin >> carouselPosts;
```

```
return (staticPosts * 1000) +
```

```
    (reelPosts * 2500) +
```

```
    (carouselPosts * 1800);
```

```
}
```

```
float calculateHandlingFee(float adBudget)
```

```
{
```

```
    if(adBudget < 50000)
```

```
        return adBudget * 0.05;
```

```
    else if(adBudget <= 100000)
```

```
        return adBudget * 0.08;
```

```
    else
```

```
        return adBudget * 0.10;
```



```
}
```

```
float calculateExtraCharges()
```

```
{
```

```
    if(campaignDays > 30)
```

```
        return (campaignDays - 30) * 500;
```

```
    return 0;
```

```
}
```

```
float calculateDiscount(float total)
```

```
{
```

```
    if(businessType == "Small" || businessType == "small")
```

```
        return total * 0.05;
```

```
    else if(businessType == "Medium" || businessType == "medium")
```

```
        return total * 0.08;
```

```
    else
```

```
        return total * 0.10;
```

```
}
```

```
float calculateGST(float total)
```

```
{
```

```
    return total * 0.16;
```

```
}
```

```
void calculateCampaignCost()
{
    if(selectedPlatform == 0)
    {
        cout << "Please Select Platform First!\n";
        return;
    }

    float postCost = calculatePostDesignCost();

    float adBudget;

    cout << "Enter Advertisement Budget: ";
    cin >> adBudget;

    float handlingFee = calculateHandlingFee(adBudget);

    float extraCharges = calculateExtraCharges();

    float platformFee =
        platformCharges[selectedPlatform - 1];

    float subTotal =
        platformFee +
        postCost +
```

**adBudget +
handlingFee +
extraCharges;**

float gst = calculateGST(subTotal);

float discount = calculateDiscount(subTotal);

**float finalBill =
subTotal +
gst -
discount;**

cout << "\n===== SOCIAL MEDIA CAMPAIGN BILL =====\n";

**cout << "Client Name: "
 << clientName << endl;**

**cout << "Business Name: "
 << businessName << endl;**

**cout << "Business Type: "
 << businessType << endl;**

**cout << "Selected Platform: "
 << platforms[selectedPlatform - 1] << endl;**


```
cout << "Campaign Duration: "
```

```
<< campaignDays
```

```
<< " Days\n";
```

```
cout << "\nPlatform Management Charges: Rs. "
```

```
<< platformFee;
```

```
cout << "\nPost Design Cost: Rs. "
```

```
<< postCost;
```

```
cout << "\nAd Budget: Rs. "
```

```
<< adBudget;
```

```
cout << "\nAd Handling Fee: Rs. "
```

```
<< handlingFee;
```

```
cout << "\nExtra Duration Charges: Rs. "
```

```
<< extraCharges;
```

```
cout << "\nGST: Rs. "
```

```
<< gst;
```

```
cout << "\nDiscount: Rs. "
```

```
<< discount;
```

```

cout << "\n-----";

cout << "\nFinal Campaign Cost: Rs. "
    << finalBill;

cout <<
"\n=====\\n";
}

void displayClientDetails()
{
    cout << "\nClient Name: "
        << clientName;

    cout << "\nBusiness Name: "
        << businessName;

    cout << "\nBusiness Type: "
        << businessType;

    cout << "\nCampaign Duration: "
        << campaignDays
        << " Days\\n";
}

int main()

```



```
{  
  
    registerClient();  
  
  
    int choice;  
  
  
    do  
    {  
        cout << "\n===== SOCIAL MEDIA MENU =====\n";  
        cout << "1. View Platforms\n";  
        cout << "2. Select Platform\n";  
        cout << "3. Calculate Campaign Cost\n";  
        cout << "4. View Client Details\n";  
        cout << "5. Exit\n";  
  
  
        cout << "Enter Choice: ";  
        cin >> choice;  
  
  
        switch(choice)  
        {  
            case 1:  
                displayPlatforms();  
                break;  
  
            case 2:  
                selectPlatform();  
                break;
```

case 3:

calculateCampaignCost();

break;

case 4:

displayClientDetails();

break;

case 5:

cout << "Program Ended.\n";

break;

default:

cout << "Invalid Choice!\n";

}

} while(choice != 5);

return 0;

}

Program 06

#include<iostream>

#include<string>

using namespace std;

```
string username, password;
```

```
string titles[20];
```

```
string contents[20];
```

```
int totalEntries = 0;
```

```
void registerUser()
```

```
{
```

```
    cout << "\n===== REGISTER =====\n";
```

```
    cout << "Enter Username: ";
```

```
    cin >> username;
```

```
    cout << "Enter Password: ";
```

```
    cin >> password;
```

```
    cout << "\nRegistration Successful!\n";
```

```
}
```

```
bool login()
```

```
{
```

```
    string user, pass;
```

```
    cout << "\n===== LOGIN =====\n";
```

```
    cout << "Enter Username: ";
```



```
cin >> user;
```

```
cout << "Enter Password: ";
```

```
cin >> pass;
```

```
if(user == username&&pass == password)
```

```
{
```

```
    cout << "\nLogin Successful!\n";
```

```
    return true;
```

```
}
```

```
cout << "\nInvalid Username or Password!\n";
```

```
return false;
```

```
}
```

```
void createEntry()
```

```
{
```

```
    if(totalEntries >= 20)
```

```
    {
```

```
        cout << "\nDiary Full!\n";
```

```
        return;
```

```
    }
```

```
cin.ignore();
```

```
cout << "\nEnter Title: ";
```

```
getline(cin, titles[totalEntries]);
```

```
cout << "Enter Content: ";
```

```
getline(cin, contents[totalEntries]);
```

```
totalEntries++;
```

```
cout << "\nEntry Added Successfully!\n";
```

```
}
```

```
void viewAllEntries()
```

```
{
```

```
if(totalEntries == 0)
```

```
{
```

```
    cout << "\nNo Entries Found!\n";
```

```
    return;
```

```
}
```

```
cout << "\n===== ALL ENTRIES =====\n";
```

```
for(int i = 0; i < totalEntries; i++)
```

```
{
```

```
    cout << i + 1 << ". "
```

```
        << titles[i] << endl;
```

```
}
```

```
}
```

```
void readFullEntry()
{
    int entryNo;

    cout << "\nEnter Entry Number: ";
    cin >> entryNo;

    if(entryNo < 1 || entryNo > totalEntries)
    {
        cout << "Invalid Entry Number!\n";
        return;
    }

    cout << "\nTitle: "
        << titles[entryNo - 1];

    cout << "\nContent: "
        << contents[entryNo - 1]
        << endl;
}

void editEntry()
{
    int entryNo;
```



```
cout << "\nEnter Entry Number To Edit: ";
```

```
cin >> entryNo;
```

```
if(entryNo < 1 || entryNo > totalEntries)
```

```
{
```

```
    cout << "Invalid Entry Number!\n";
```

```
    return;
```

```
}
```

```
cin.ignore();
```

```
cout << "Enter New Title: ";
```

```
getline(cin, titles[entryNo - 1]);
```

```
cout << "Enter New Content: ";
```

```
getline(cin, contents[entryNo - 1]);
```

```
cout << "\nEntry Updated Successfully!\n";
```

```
}
```

```
void deleteEntry()
```

```
{
```

```
    int entryNo;
```

```
    cout << "\nEnter Entry Number To Delete: ";
```

```
    cin >> entryNo;
```

```
if(entryNo < 1 || entryNo > totalEntries)
```

```
{
```

```
    cout << "Invalid Entry Number!\n";
```

```
    return;
```

```
}
```

```
for(int i = entryNo - 1; i < totalEntries - 1; i++)
```

```
{
```

```
    titles[i] = titles[i + 1];
```

```
    contents[i] = contents[i + 1];
```

```
}
```

```
totalEntries--;
```

```
cout << "\nEntry Deleted Successfully!\n";
```

```
}
```

```
void deleteAllEntries()
```

```
{
```

```
    totalEntries = 0;
```

```
    cout << "\nAll Entries Deleted Successfully!\n";
```

```
}
```

```
int main()
```



```
{  
  
    registerUser();  
  
    if(!login())  
        return 0;  
  
    int choice;  
  
    do  
    {  
        cout << "\n===== PERSONAL STUDENT DIARY =====\n";  
        cout << "1. Create New Diary Entry\n";  
        cout << "2. View All Entries\n";  
        cout << "3. Read Full Entry\n";  
        cout << "4. Edit Entry\n";  
        cout << "5. Delete One Entry\n";  
        cout << "6. Delete All Entries\n";  
        cout << "7. Exit\n";  
  
        cout << "Enter Choice: ";  
        cin >> choice;  
  
        switch(choice)  
        {  
            case 1:  
                createEntry();
```

break;

case 2:

viewAllEntries();

break;

case 3:

readFullEntry();

break;

case 4:

editEntry();

break;

case 5:

deleteEntry();

break;

case 6:

deleteAllEntries();

break;

case 7:

cout << "\nProgram Ended.\n";

break;

default:

cout << "\nInvalid Choice!\n";

}

} while(choice != 7);

return 0;

}